# Towards an Approach for Classifying Skin Lesions Using Convolutional Neural Networks

Rodrigo Guedes de Souza[1], Fabricio Ourique[2], Analúcia Schiaffino Morales[1], Antonio Carlos Sobieranski[1], and Alison R. Panisson[1]

[1]*Federal University of Santa Catarina (UFSC) - Brazil*

[2]*Queen Mary University of London - United Kingdom*

*guedes.rodrigo@grad.ufsc.br, f.ourique@qmul.ac.uk, {analucia.morales,a.sobieranski,alison.panisson}@ufsc.br*

Keywords: Computer Vision, Convolutional Neural Networks, Skin Lesion

Abstract: Early detection of skin cancer is a critical challenge, particularly in regions with limited access to dermatologists. This paper presents a deep learning approach using Convolutional Neural Networks (CNNs) to classify skin lesions as benign or malignant. We developed a preprocessing and augmentation pipeline tailored for medical images to enhance model generalization. Our model is based on a modified ResNet18 architecture with attention mechanisms, implemented using TensorFlow and Keras. Additionally, we explore explainability techniques to provide interpretable visual explanations that support the model's predictions. Trained on the International Skin Imaging Collaboration (ISIC) dataset, the proposed approach achieved an accuracy of 99.42% on the test set and 88.12% on the validation set. These results highlight the potential of CNN-based methods in assisting early skin cancer detection, particularly in automated screening scenarios where expert evaluation may be limited.

## 1 INTRODUCTION

Cancer is characterized by disordered cellular growth that can affect various organs and tissues, resulting in significant health damage. In the case of skin, this process mainly occurs in cells that begin to multiply abnormally and form lesions (Apalla et al., 2017). These lesions can be either benign, such as nevi (moles) without any risk of progression, or malignant, such as melanoma and other carcinomas that may invade surrounding tissues and potentially spread to other parts of the body (Paolino et al., 2017). In the context of skin cancer, early detection is crucial, as it facilitates treatment and increases the chances of cure.

The early detection of potentially malignant skin lesions represents a critical challenge in medicine, primarily due to the wide variety of clinical presentations and the imbalance between benign and malignant cases. Recent studies show that, in addition to clinical and histopathological examination techniques (Calin et al., 2013), computational approaches based on Convolutional Neural Networks (CNNs) (Tang et al., 2020; Dildar et al., 2021) can provide reliable support for the diagnosis of skin cancer (Shetty et al., 2022). Early research in this area has explored various architectures such as ResNet (Alruwaili and Mohamed, 2025) and EfficientNets (Ali et al., 2022), obtaining promising results on public datasets such as the one provided by the International Skin Imaging Collaboration (ISIC).

The present work proposes an approach for classifying skin lesions into benign or malignant categories using a Convolutional Neural Network (CNN) trained on the ISIC skin image dataset. This classification model is supported by a dedicated preprocessing pipeline and data augmentation techniques specifically tailored for medical imaging, aiming to enhance generalization and robustness. To align with the critical requirements of healthcare applications, particularly the need for systems that assist healthcare professionals in decision-making, we integrate an interpretability component into the proposed pipeline. Specifically, we employ Gradient-weighted Class Activation Mapping (Grad-CAM) to generate visual explanations in the form of heatmaps. These heatmaps highlight the regions of the lesion that most strongly influence the model's classification, thereby offering transparency and aiding clinicians in understanding and trusting the model's output. Preliminary results demonstrate promising performance, with an AUC-ROC of 0.92, a specificity of 0.96, and a validation accuracy of 88.12%. These findings underscore the

potential of the proposed approach to support early diagnosis and preliminary screening of skin cancer, particularly in settings with limited access to dermatology specialists.

## 2 BACKGROUND

Machine learning techniques for image classification have demonstrated great potential in extracting and analyzing complex visual patterns in a wide range of domains, including medical imaging (Li et al., 2024). In recent years, deep learning models, especially CNNs, have emerged as state-of-the-art solutions for tasks where subtle differences in image textures, colors, and shapes are critical for accurate diagnosis (Gu et al., 2018). In the context of skin lesion analysis, CNNs enable the automated detection and classification of lesions into categories such as benign, malignant, and indeterminate (Tschandl et al., 2020), which is essential for early skin cancer screening.

CNNs are composed of several key types of layers. Convolutional layers are used to extract local features by applying learnable filters across the input images. Pooling layers follow to reduce the spatial dimensionality of the feature maps, thus decreasing the total number of parameters and computational cost, while preserving the most salient information. Fully connected layers then integrate these learned features to perform the final classification (Alzubaidi et al., 2021). Additionally, modern CNNs architectures incorporate useful techniques such as batch normalization, dropout, and other forms of regularization. These methods help stabilize and accelerate training, while also reducing the risk of overfitting. As a result, CNNs are particularly effective in capturing hierarchical features, where early layers identify simple visual cues (e.g., edges and textures) and later layers discern higher-level structures critical to differentiating skin lesions (Zeiler and Fergus, 2014).

The availability of large, publicly accessible datasets, such as the International Skin Imaging Collaboration (ISIC) dataset[1], has been instrumental in advancing research in computer-aided skin lesion classification. These datasets provide a diverse range of dermatoscopic images that facilitate transfer learning and robust feature extraction without the need for extensive domain-specific feature engineering. Consequently, CNN-based systems have shown promising results in accurately differentiating between benign lesions and malignant tumors, even in the presence of significant intra-class variability and challenging imaging conditions (Traore et al., 2018).

Given the high stakes involved in skin cancer diagnosis, wherein early detection is crucial for effective treatment, leveraging CNNs to provide automated support to clinicians represents a significant step forward in medical imaging. In this work, we contrast traditional image processing techniques with modern CNN architectures, discuss challenges such as class imbalance and data augmentation, and outline the benefits of incorporating interpretability modules (such as Grad-CAM) to provide visual explanations that can support clinical decision-making.

## 3 RELATED WORK

In (Alruwaili and Mohamed, 2025), the authors proposed a fusion-level deep learning model for skin disease diagnosis by combining EfficientNet-B0, EfficientNet-B2, and ResNet50 in independent branches. The fused features, processed by dense and dropout layers, led to an accuracy of 99.14% on the Kaggle Skin Diseases Image Dataset (27,153 images, split 80/10/10 for train/validation/test), with good precision, recall, and F1-score.

In (Almufareh et al., 2024), the authors show that recent advancements in AI have propelled skin cancer diagnostics. This study developed a robust CNN model using dermatoscopic images from public datasets (DermIS, DermQuest, DermIS&Quest, and ISIC2019) for early melanoma detection. Leveraging transfer learning, the model extracts and classifies relevant features using convolutional and pooling layers, followed by fully connected layers. Evaluations revealed improved detection accuracy, demonstrating the model's strong discriminative ability and its potential to support clinical decision-making for early melanoma diagnosis.

In (Ali et al., 2022), the authors implemented an image preprocessing pipeline that removes artifacts (e.g., hairs), augments the dataset, and normalizes all images to a standard size. Applied Transfer learning using EfficientNet variants fine-tuned on the HAM10000 dataset. The best model, EfficientNet B4, achieved an F1 score of 87% and a Top-1 accuracy of 87.91%.

In (Mamadapur et al., 2024), the authors propose a nine-class skin lesion classifier based on convolutional neural networks, utilizing the DenseNet architecture. They employ transfer learning by initializing pre-trained weights derived from a large-scale image dataset. Additionally, the authors integrate multiple datasets and apply various data augmentation tech-

[1] https://www.isic-archive.com

niques to enhance data diversity. Their results demonstrate an accuracy of 96.95%, leading to the conclusion that data augmentation effectively improves the diversity of the training data and, consequently, enhances the resulting model's performance.

In (Cino et al., 2025), the authors address skepticism regarding automatic skin lesion classification by artificial intelligence through the proposal of explainable approaches. They investigate accurate AI-based algorithms for skin lesion classification, exploring six CNN-based architectures that also provide visual explanations to enhance trust and confidence in these diagnostic tools. To achieve explainability, they utilize t-distributed Stochastic Neighbor Embedding (t-SNE) and Gradient-weighted Class Activation Mapping (Grad-CAM) to offer insights into how the model makes classifications, thus improving interpretability.

While inspired by these works, we employed a different CNN architecture, preprossesing and augmentation techniques, combined with an explainer, to develop our approach for skin lesion classification, achieving competitive results, as highlighted in Section 5.

# 4 CNN-BASED CLASSIFIER FOR SKIN LESIONS

In this work, we design and train CNN-based classifiers to recognize and categorize skin lesions into two classes: *benign* or *malignant*. The classifier architecture was developed based on a transfer learning approach (West et al., 2007) utilizing the ResNet18 network pre-trained on the ImageNet dataset. This choice allowed the model to begin with a robust base for visual feature extraction, which is critical in analyzing skin lesion images where subtle textural and morphological differences determine whether a tumor is benign or malignant.

To adapt ResNet18 to the specific task of skin lesion classification, its final fully connected layer was replaced with a new linear layer that outputs a single value. This output is subsequently passed through a sigmoid activation function, enabling the use of the Binary Cross-Entropy With Logits Loss (BCEWithLogitsLoss). To further address the inherent class imbalance in the dataset (where benign lesions significantly outnumber malignant ones) the loss function is weighted by specifying a `pos_weight` parameter that increases the importance of the malignant class during training.

Furthermore, the training process is executed on a GPU whenever available, ensuring accelerated computation. This is particularly important for CNN-based models, as the GPU significantly reduces the time required for complex convolutional operations and gradient calculations. The implementation leverages the PyTorch framework to manage device transfers, perform convolutional operations, and update model parameters using the *Adam optimizer* configured with a low learning rate and weight decay to prevent overfitting.

In addition, the data pipeline incorporates a careful separation of the training and validation sets. The training data is organized into subdirectories corresponding to benign and malignant lesions, while a distinct validation set is maintained for performance evaluation. This separation, along with the application of data augmentation techniques (applied selectively to the malignant class in other parts of the project), ensures that the model learns effectively from a balanced representation of the classes.

In summary, the implementation combines the powerful feature extraction capabilities of a pre-trained ResNet18 with architectural modifications and class-balancing strategies. The replacement of the final layer, the use of a weighted loss function to focus on the minority class, and the careful organization of the training and validation data collectively contribute to a robust and efficient system for skin lesion classification, suitable for clinical decision support.

## 4.1 Dataset and Data Augmentation

The developed system utilizes the ISIC 2018 dataset, which provides 12279 high-quality dermatoscopic images of skin lesions. Given the inherent imbalance in medical imaging datasets (with benign lesions significantly outnumbering malignant ones) the dataset was processed to enhance the representation of the malignant class. In our study, the original dataset was expanded through a combination of data augmentation techniques and careful splitting to form a robust training and testing set. Specifically, while the benign class was initially composed of approximately 7186 images, the malignant class originally had around 1230 images. To address this disparity, a custom data augmentation pipeline was implemented that generates multiple augmented copies of each malignant image via randomized geometric and photometric transformations.

The augmentation includes random rotations, horizontal and vertical flips, affine transformations, brightness-contrast adjustments, and elastic deformations, all applied exclusively to the malignant cases. Also, the indeterminate class contained 2634 images that remained non-determinate after the preprocess-

ing step. Fig. 1 depicts examples of an augmented image with a horizontal flip (on the right) and a processed image with brightness-contrast adjustment (on the left).

As a result, the effective number of malignant images in the training set was increased substantially, yielding a final count of about 6,200 malignant samples. Combined with the benign examples and after applying the augmentation to all training samples, the training dataset was expanded to a total of 13,386 images for training. A separate testing (validation) set of 1,663 images was maintained, comprising untreated (or minimally processed) images to accurately evaluate the model's generalization.

This approach increases the representation of malignant cases during training, also preserves the original characteristics of the images in the testing set, thereby creating a robust framework for evaluation. The data augmentation pipeline was developed using the Albumentations library[2], which allows for a diverse range of reproducible transformations. By doing so, the model is better equipped to learn and recognize subtle, clinically relevant features in malignant lesions. This improved learning process ultimately enhances the sensitivity and overall performance of the classifier in real-world skin lesion analysis.

## 4.2 Pre-processing

The pre-processing pipeline is a critical component of our approach for skin lesion classification, ensuring that the input images are standardized and optimized for feature extraction by the CNN. All dermatoscopic images are organized into separate directories corresponding to their classes (benign and malignant) prior to processing. For both training and validation, images are uniformly resized to $299 \times 299$ pixels. This resizing not only facilitates batch processing in the network but also conforms to the dimensions used in the pre-training of the adopted model.

For the training set, additional randomized transformations are applied to improve the model's robustness. Specifically, transformations such as random horizontal and vertical flips, random rotations up to 20 degrees, and random translations and scaling are incorporated using the TorchVision library[3]. These augmentations introduce variability in orientation and illumination, which are common challenges in dermatoscopic imaging. Furthermore, the training pipeline includes normalization of pixel intensities using the mean and standard deviation derived from ImageNet, ensuring consistency with the pre-trained ResNet18's expected input distribution.

Given the substantial class imbalance (with benign lesions outnumbering malignant ones) the system employs a custom data augmentation strategy for the malignant class. In a dedicated dataset class, each malignant image is augmented multiple times using the Albumentations library to generate diversified versions. This augmentation process involves a sequence of transformations (including random rotations, horizontal and vertical flips, affine transformations, brightness-contrast adjustments, and elastic distortions) applied exclusively to malignant images. Such an approach effectively increases the representation of the malignant class in the training set, allowing the model to learn more robust features specific to these critical cases.

The overall pre-processing pipeline not only standardizes image dimensions and intensity distributions but also minimizes unwanted variability arising from differences in background, lighting, and noise. By combining conventional transformations with targeted data augmentation for malignant lesions, the system ensures that the CNN is exposed to a balanced and enriched set of features, thereby enhancing its ability to discern subtle differences between benign and malignant skin lesions.

## 4.3 Model Training

To perform skin lesion classification, we adapted a pre-trained ResNet18 architecture from the PyTorch framework. The original ResNet18, originally trained on ImageNet (Deng et al., 2009), serves as a powerful feature extractor due to its residual structure, which effectively mitigates the vanishing gradient problem in deep networks. In our implementation, the final fully connected layer of ResNet18 was replaced with a new linear layer that outputs a single scalar. This scalar is subsequently passed through a sigmoid function, enabling the network to output a probability value for each lesion, then used to make a binary decision between benign and malignant.

Given the natural imbalance in the dataset, with approximately 7,200 benign images and 1,200 malignant images, the training strategy incorporates two key mechanisms to address this challenge. First, the loss function used is BCEWithLogitsLoss, which was modified by using a higher positive weight. This weighted loss emphasizes errors on malignant cases and thus drives the network to focus on the critical features of the minority class. Second, the data loading pipeline employs a custom dataset class that applies aggressive data augmentation selectively to ma-

[2]https://albumentations.ai/

[3]https://pytorch.org/vision/

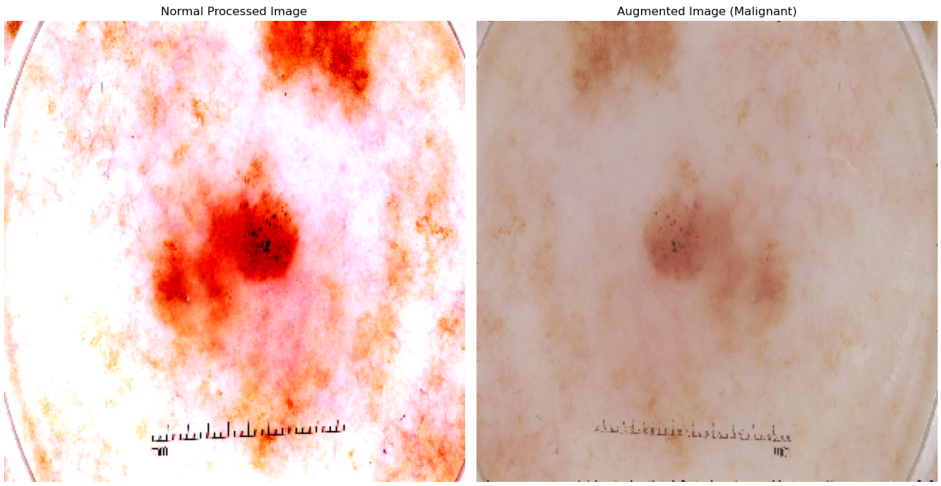


Figure 1: Comparative between a Normal Processed Image and a Augmented Image (Malignant class image).

lignant images. This augmentation pipeline uses a variety of transformations (random rotations, flips, affine deformations, brightness and contrast adjustments, and elastic distortions) to generate multiple augmented versions of each malignant image.

The training process capitalizes on GPU acceleration using PyTorch's device management. All model parameters, as well as input batches, are explicitly transferred to the GPU (if available) to expedite the computational process. The training loop involves iterating through mini-batches, computing the weighted binary cross-entropy loss, and updating the model weights using the Adam optimizer configured with a low learning rate (0.0001) and a weight decay term to prevent overfitting. At the end of each epoch, the model's performance is evaluated on a separate validation dataset. Both training and validation losses and accuracies are recorded, and these metrics are printed in a clear, tabulated format for each epoch.

A key aspect of the training protocol is the continuous monitoring of validation performance. After each epoch, if the validation accuracy improves compared to previous epochs, the model's checkpoint is saved, including the epoch number, model state, optimizer state, and performance metrics. This systematic checkpointing ensures that the best performing model is preserved for further evaluation and deployment. Table 1 contains all training Epochs and their respective Metrics.

Table 1: Accuracy Evolution during Transfer Learning.

| Epoch | Loss | Acc. | Val_loss | Val_acc |
|---|---|---|---|---|
| 1 | 0.2200 | 95.30% | 0.8212 | 79.25% |
| 2 | 0.0871 | 98.16% | 1.1275 | 86.49% |
| 3 | 0.0611 | 98.75% | 1.2900 | 86.49% |
| 4 | 0.0523 | 98.97% | 2.0527 | 85.68% |
| 5 | 0.0448 | 99.12% | 1.2344 | 86.09% |
| 6 | 0.0338 | 99.42% | 1.6744 | 88.12% |
| 7 | 0.0325 | 99.38% | 2.0619 | 85.68% |
| 8 | 0.0294 | 99.46% | 2.2287 | 84.78% |
| 9 | 0.0257 | 99.50% | 4.4896 | 77.54% |
| 10 | 0.0240 | 99.54% | 3.1305 | 83.81% |
| 11 | 0.0239 | 99.53% | 2.1785 | 86.66% |
| 12 | 0.0221 | 99.60% | 1.4269 | 84.05% |
| 13 | 0.0175 | 99.73% | 2.7681 | 86.17% |
| 14 | 0.0206 | 99.64% | 2.1544 | 83.48% |
| 15 | 0.0174 | 99.68% | 3.4936 | 84.95% |
| 16 | 0.0140 | 99.75% | 2.3945 | 83.56% |
| 17 | 0.0137 | 99.78% | 3.2623 | 84.05% |
| 18 | 0.0151 | 99.73% | 3.4501 | 85.03% |
| 19 | 0.0152 | 99.74% | 1.9122 | 84.95% |
| 20 | 0.0123 | 99.81% | 2.9105 | 85.60% |

## 5 EVALUATION

The evaluation of our skin lesion classifier was based on several key metrics that together provide a comprehensive view of its performance. We used the AUC-ROC to assess the model's discriminative power between benign and malignant lesions, specificity and Negative Predictive Value (NPV) to ensure benign cases are accurately identified, and sensitivity to measure the detection rate of malignant lesions. In addition, we monitored training and validation accuracies to gauge potential overfitting.

Fig. 2 shows the ROC-Curve for the classification model. The model achieves an AUC-ROC of 0.8715. AUC-ROC measures the model's ability to

distinguish between classes across varied thresholds, and values closer to 1 indicate excellent discrimination. In our case, this value demonstrates that the model is highly capable of differentiating between malignant and benign lesions.

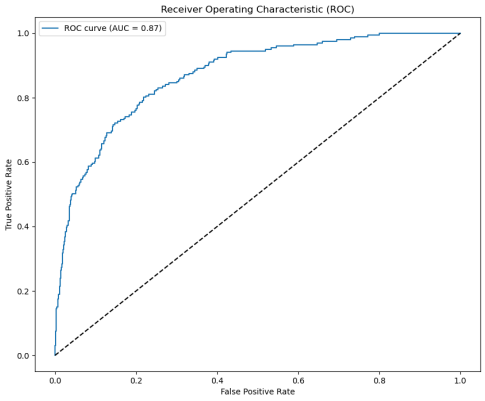


Figure 2: ROC-Curve for the classification model.

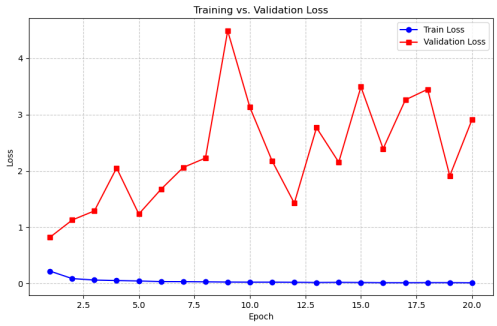


Figure 3: Train and Validation Loss Evolution.

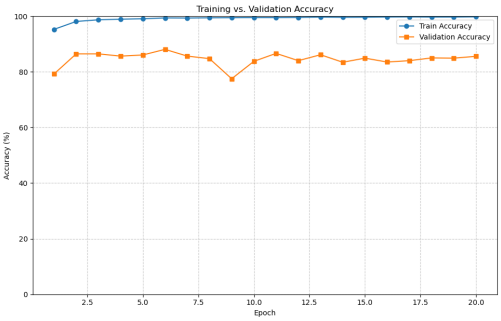


Figure 4: Accuracy Evolution.

Specificity is another strong point of the model, reaching 94.94%. This metric reflects the model's effectiveness in correctly identifying benign lesions and minimizing false positives. High specificity is crucial in medical diagnostics to avoid unnecessary anxiety or interventions. The Negative Predictive Value (NPV) is 90.88%, meaning that when the model predicts a lesion is benign, there is a high probability that it truly is benign. This further reinforces the model's reliability in ruling out malignancies.

Fig. 3 shows the training and validation loss evolution, and Fig. 4 shows the accuracy evolution of the model. As observed, during training, the model achieved an accuracy of 99.42%, indicating that it fits the training data very well. However, the validation accuracy dropped to 88.12%, which is a sign of potential overfitting. This discrepancy suggests that while the model learns the training patterns almost perfectly, its performance on unseen data is lower.

## 5.1 Grad-CAM support

Grad-CAM (Gradient-weighted Class Activation Mapping) is an interpretability technique that provides visual explanations for the predictions of convolutional neural networks (Selvaraju et al., 2020). By highlighting regions in an input image that most strongly influence the model's decision, Grad-CAM enhances the understanding of the model's focus during prediction.

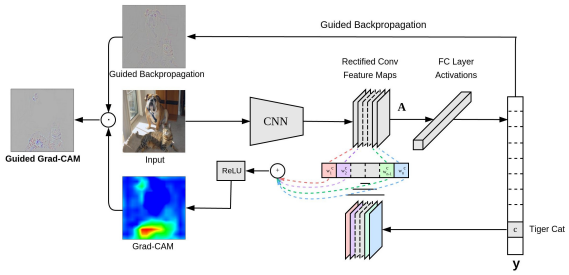


Figure 5: Grad CAM general architecture (Selvaraju et al., 2020).

The original Grad-CAM workflow, illustrated in Fig. 5, integrates feature maps from a convolutional layer with gradient-based importance weights corresponding to a specific class (e.g., "tiger cat"). These weighted feature maps are combined and passed through a ReLU function to produce the Grad-CAM heatmap. In parallel, the input image undergoes guided backpropagation—a technique that traces fine-grained features responsible for neuron activation. The final Guided Grad-CAM visualization is obtained by performing element-wise multiplication of the Grad-CAM heatmap with the guided backpropagation output, resulting in a high-resolution image that merges class-discriminative localization with detailed spatial structure. Fig. 7 presents two examples of heatmaps and Guided Grad-CAM results generated for different classes from the same input image.

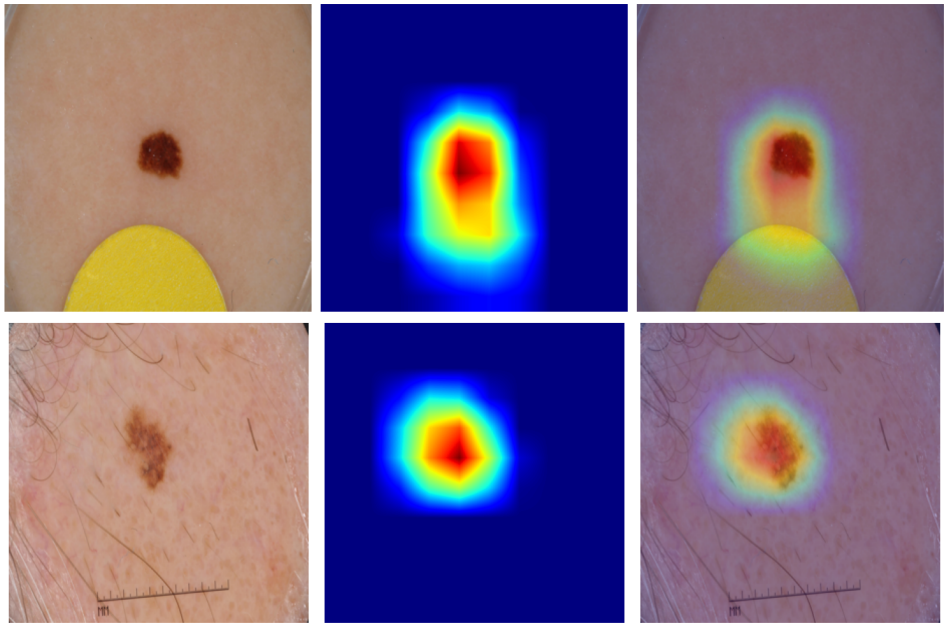

Figure 6: Grad-CAM heatmaps for two benign tumor examples. Concentration of heat in the lesion's area indicates that the model is evaluating the correct location.

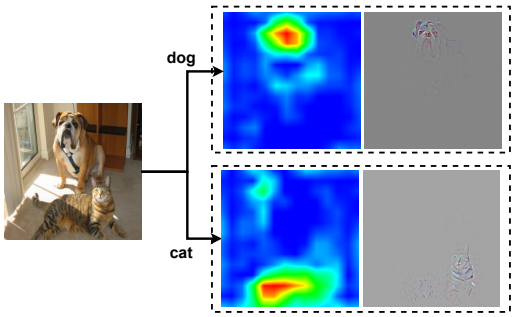


Figure 7: Example of Heatmaps and Guided Grad-CAM generated for different classes (Selvaraju et al., 2020).

In the context of skin lesion classification, this interpretability is particularly valuable, as it enables clinicians to verify whether the model is focusing on relevant pathological features rather than extraneous background information. Such visual cues not only enhance trust in the model by making its decision-making process more transparent but also aid in detecting potential errors in its behavior. Fig. 6 illustrates the application of Grad-CAM to images used in this study.

## 6 CONCLUSION

In this work, we presented a deep learning approach for the classification of skin lesions using convolutional neural networks. Our approach leverages transfer learning from a pre-trained ResNet18 model, which was adapted by replacing its final fully connected layer to output a single probability score for skin lesion malignancy. To address the inherent class imbalance in our dataset, we employed a two-pronged strategy: a weighted loss function emphasizing the minority (malignant) class and a specialized data augmentation pipeline that generates multiple transformed copies of malignant images. These techniques enabled our model to achieve a high overall discriminative performance, as evidenced by an AUC-ROC of 0.8715 and good specificity (94.94%) and negative predictive value (90.88%), ensuring that false positives are kept to a minimum.

However, despite a high training accuracy of 99.42%, the validation accuracy dropped to 88.12% and the sensitivity remained relatively low at 51.24%. This gap suggests that while the model is capable of correctly classifying benign lesions, it struggles with detecting malignant ones — a critical shortcoming given the clinical importance of early cancer detection. The observed performance, particularly the imbalance between sensitivity and specificity, underscores the challenges associated with classifying skin lesions and highlights the need for further refinement.

The results indicate several promising avenues for future work. First, further optimization of data augmentation techniques and class weighting could help boost sensitivity without sacrificing specificity. Second, exploring alternative architectures or ensemble

methods might lead to a more balanced performance across all classes. Third, integrating interpretability tools, such as Grad-CAM, into the proposed approach offers clinicians visual insights into how the model is making decisions, thereby increasing trust and assisting in diagnostic assessments. Combining the classifier with the visual explainer, the proposed approach serves as a valuable tool for enhancing decision-making support.

Overall, our study demonstrates the potential of CNN-based approaches to provide valuable support for early skin cancer diagnosis, contributing to the broader effort of optimizing computer-aided diagnostic systems in healthcare. Future research should focus on addressing the limitations in sensitivity and further validating the model on clinically diverse datasets to ensure its robustness and generalizability in real-world applications.

# REFERENCES


Ali, K., Shaikh, Z. A., Khan, A. A., and Laghari, A. A. (2022). Multiclass skin cancer classification using efficientnets – a first step towards preventing skin cancer. *Neuroscience Informatics*, 2(4):100034.

Almufareh, M. F., Tariq, N., Humayun, M., and Khan, F. A. (2024). Melanoma identification and classification model based on fine-tuned convolutional neural network. *DIGITAL HEALTH*, 10:20552076241253757.

Alruwaili, M. and Mohamed, M. (2025). An integrated deep learning model with efficientnet and resnet for accurate multi-class skin disease classification. *Diagnostics*, 15(5).

Alzubaidi, L., Zhang, J., Humaidi, A. J., Al-Dujaili, A., Duan, Y., Al-Shamma, O., Santamaría, J., Fadhel, M. A., Al-Amidie, M., and Farhan, L. (2021). Review of deep learning: concepts, cnn architectures, challenges, applications, future directions. *Journal of Big Data*, 8(1). Cited by: 4083; All Open Access, Gold Open Access, Green Open Access.

Apalla, Z., Nashan, D., Weller, R. B., and Castellsagué, X. (2017). Skin cancer: Epidemiology, disease burden, pathophysiology, diagnosis, and therapeutic approaches. *Dermatology and Therapy*, 7(1):5–19.

Calin, M. A., Parasca, S. V., Savastru, R., Calin, M. R., and Dontu, S. (2013). Optical techniques for the noninvasive diagnosis of skin cancer. *Journal of Cancer Research and Clinical Oncology*, 139(7):1083–1104.

Cino, L., Distante, C., Martella, A., and Mazzeo, P. L. (2025). Skin lesion classification through test time augmentation and explainable artificial intelligence. *Journal of Imaging*, 11(1):15.

Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. (2009). Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255.

Dildar, M., Akram, S., Irfan, M., Khan, H. U., Ramzan, M., Mahmood, A. R., Alsaiari, S. A., Saeed, A. H. M., Alraddadi, M. O., and Mahnashi, M. H. (2021). Skin cancer detection: A review using deep learning techniques. *International Journal of Environmental Research and Public Health*, 18(10).

Gu, J., Wang, Z., Kuen, J., Ma, L., Shahroudy, A., Shuai, B., Liu, T., Wang, X., Wang, G., Cai, J., and Chen, T. (2018). Recent advances in convolutional neural networks. *Pattern Recognition*, 77:354 – 377. Cited by: 4543; All Open Access, Green Open Access.

Li, Y., El Habib Daho, M., Conze, P.-H., Zeghlache, R., Le Boité, H., Tadayoni, R., Cochener, B., Lamard, M., and Quellec, G. (2024). A review of deep learning-based information fusion techniques for multimodal medical image classification. *Computers in Biology and Medicine*, 177:108635.

Mamadapur, P., Dhanyal, S. S., Mulla, H., and Hosur, S. (2024). Skin lesion classification using convolution neural networks with an enhanced accuracy. In *2024 International Conference on Innovation and Novelty in Engineering and Technology (INNOVA)*, volume 1, pages 1–5. IEEE.

Paolino, G., Donati, M., Didona, D., Mercuri, S. R., and Cantisani, C. (2017). Histology of non-melanoma skin cancers: An update. *Biomedicines*, 5(4).

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2020). Grad-cam: Visual explanations from deep networks via gradient-based localization. *International Journal of Computer Vision*, 128(2):336–359.

Shetty, B., Fernandes, R., Rodrigues, A. P., Chengoden, R., Bhattacharya, S., and Lakshmanna, K. (2022). Skin lesion classification of dermoscopic images using machine learning and convolutional neural network. *Scientific Reports*, 12(1):18134.

Tang, P., Liang, Q., Yan, X., Xiang, S., and Zhang, D. (2020). Gp-cnn-dtel: Global-part cnn model with data-transformed ensemble learning for skin lesion classification. *IEEE Journal of Biomedical and Health Informatics*, 24(10):2870–2882.

Traore, B. B., Kamsu-Foguem, B., and Tangara, F. (2018). Deep convolution neural network for image recognition. *Ecological Informatics*, 48:257 – 268. Cited by: 295; All Open Access, Green Open Access.

Tschandl, P., Rinner, C., Apalla, Z., Argenziano, G., Codella, N., Halpern, A., Janda, M., Lallas, A., Longo, C., Malvehy, J., Paoli, J., Puig, S., Rosendahl, C., Soyer, H. P., Zalaudek, I., and Kittler, H. (2020). Human–computer collaboration for skin cancer recognition. *Nature Medicine*, 26(8):1229–1234.

West, J., Ventura, D., and Warnick, S. (2007). Spring research presentation: A theoretical foundation for inductive transfer. *Brigham Young University, College of Physical and Mathematical Sciences*, 1(08).

Zeiler, M. D. and Fergus, R. (2014). Visualizing and understanding convolutional networks. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 8689 LNCS(PART 1):818 – 833. Cited by: 11015; All Open Access, Bronze Open Access.